Mobile Testing with Appium + Playwright TypeScript

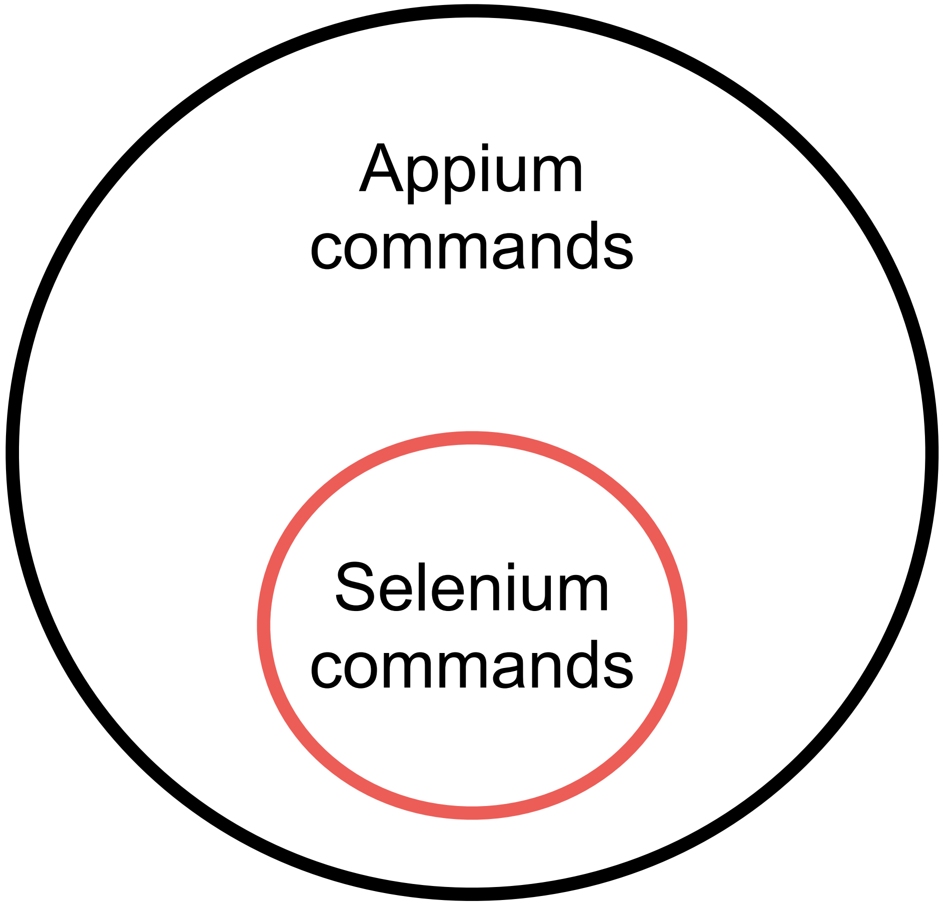
Automated Testing

https://appium.io/

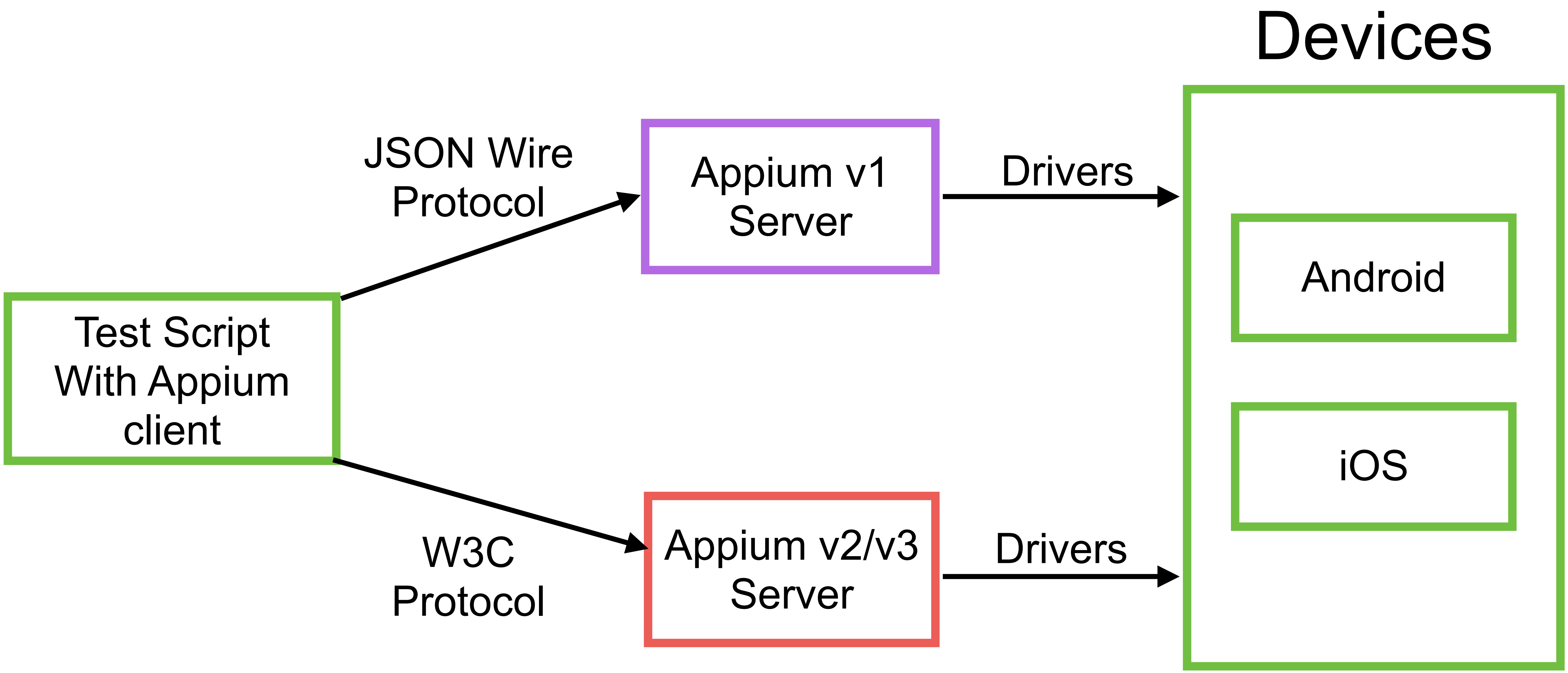
Automated Testing

Appium

Automation tool for mobile app
Selenium for mobile
High compatibility with selenium



Architecture



Appium Tools

Appium Server via npm

Appium Inspector

Android	iOS	Web	Desktop

https://appium.io/docs/en/latest/quickstart/

Appium Drivers

Platform	Driver	Platform Versions	Appium Version
iOS	XCUITest	9.3+	1.6.0+
	UIAutomation	8.0 to 9.3	All
Android	Espresso	?+	1.9.0+
	UiAutomator2	?+	1.6.0+
	UiAutomator	4.3+	All
Mac	Mac	?+	1.6.4+
Windows	Windows	10+	1.6.0+

https://appium.io/docs/en/latest/ecosystem/drivers/

Appium Clients

JavaScript/TypeScript

Python

Java

C#

Robotframework

https://appium.io/docs/en/latest/ecosystem/clients/

Let’s start with Appium

Appium doctor (android, iOS)

appium™

Appium Inspector —/wp/hub→ Appium 1 Server —driver→ Target App in device

Steps to run

Appium doctor and drivers
Start server
Provide target apps (api, ipa/app)
Appium Inspector

Check environments

Android	iOS
APPIUM_HOME	APPIUM_HOME
ANDROID_HOME	Xcode
JAVA_HOME	Xcode command line tools
adb, emulator, apkanalyzer	ios-deploy, applesimutils
NodeJS	NodeJS

List of android’s devices

$adb devices

List of iOS’s devices

$xcrun xctrace list devices

```
== Simulators ==
Apple TV Simulator (15.2) (9666637B-AD71-47CD-932D-DE7BA9096F46)
Apple TV 4K (2nd generation) Simulator (15.2) (2B6A15B3-4A03-41B5-B6C6-D7727DCA94D8)
Apple TV 4K (at 1080p) (2nd generation) Simulator (15.2) (EC5B5310-9DE7-4CB2-8AE2-172FC885DB8C)
iPad (9th generation) Simulator (15.2) (03B84395-1A2B-4759-B01A-171CD4F8F796)
iPad Air (4th generation) Simulator (15.2) (6389A3CA-7805-452D-9110-5B3796A9D9BB)
iPad Pro (11-inch) (3rd generation) Simulator (15.2) (40E522A5-14CE-453C-A54C-29D096F19236)
iPad Pro (12.9-inch) (5th generation) Simulator (15.2) (1E6E63EF-FA56-4E31-8082-13B0C664AD33)
iPad Pro (9.7-inch) Simulator (15.2) (9C7181D9-73D6-48BF-8972-80A73ADB2EA9)
iPad mini (6th generation) Simulator (15.2) (01F8F0AB-1AF5-4FBE-9587-B7713F7BBC6F)
iPhone 11 Simulator (15.2) (1E75E325-57EC-4D6D-85BF-9958B0A9B158)
iPhone 11 Pro Simulator (15.2) (7D50F81C-3521-447F-A79C-8613C6C58FEF)
iPhone 11 Pro Max Simulator (15.2) (FB336CEE-A241-43D1-8704-B0632EA9FD28)
iPhone 12 Simulator (15.2) (5325CF3D-9576-462A-A110-49A5D76665FE)
```

List of Appium driver

$appium driver list

```
✔ Listing available drivers (rerun with --verbose for more info)
- uiautomator2@7.0.0 [installed (npm)]
- xcuitest [not installed]
- espresso [not installed]
- mac2 [not installed]
- windows [not installed]
- safari [not installed]
- gecko [not installed]
- chromium [not installed]
```

https://appium.io/docs/en/latest/ecosystem/drivers/

Appium Server

Appium Server

Install via npm (required Node.js)

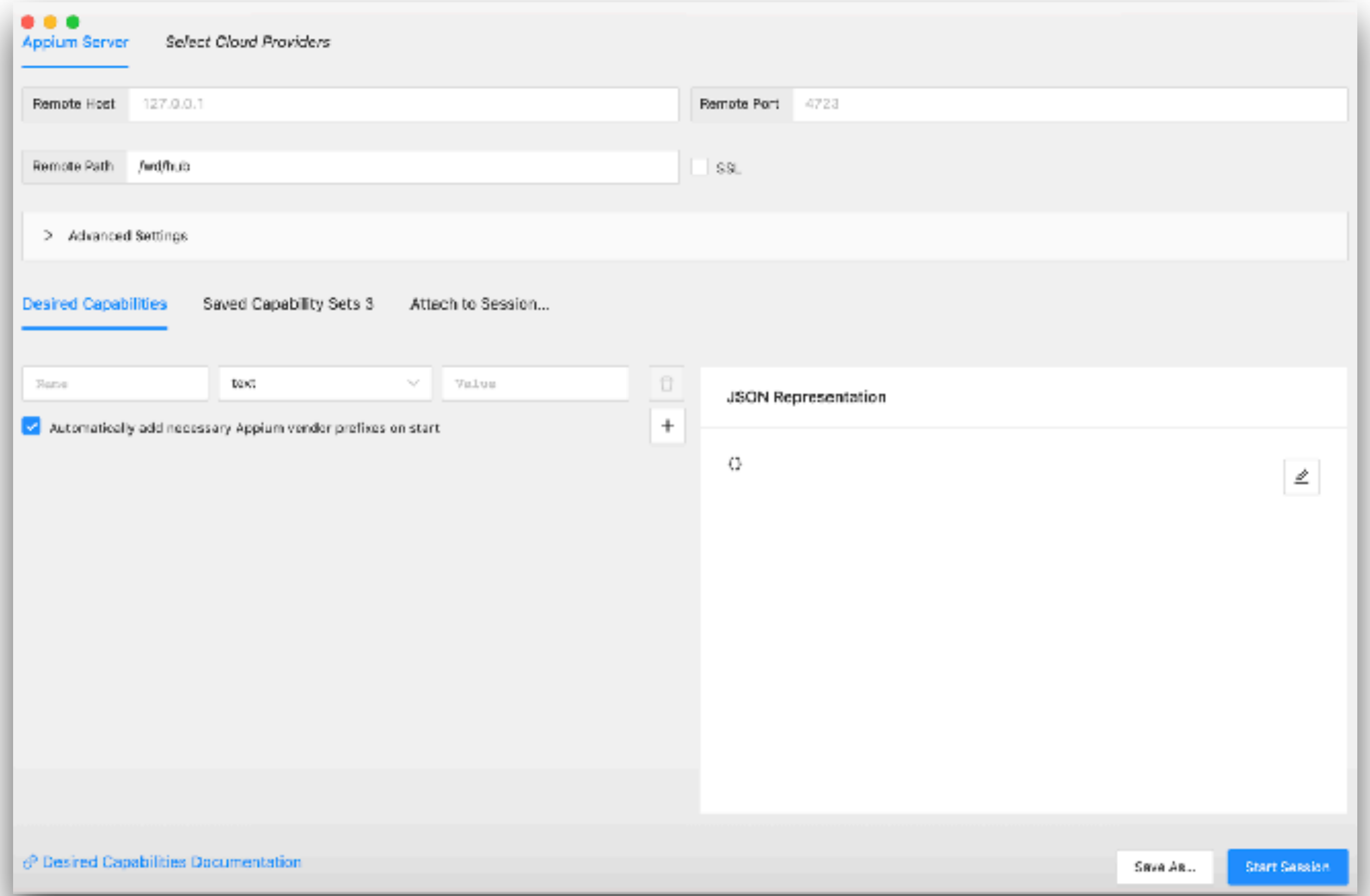
```
$npm install -g appium
$appium
```

https://github.com/appium/appium

Appium Inspector

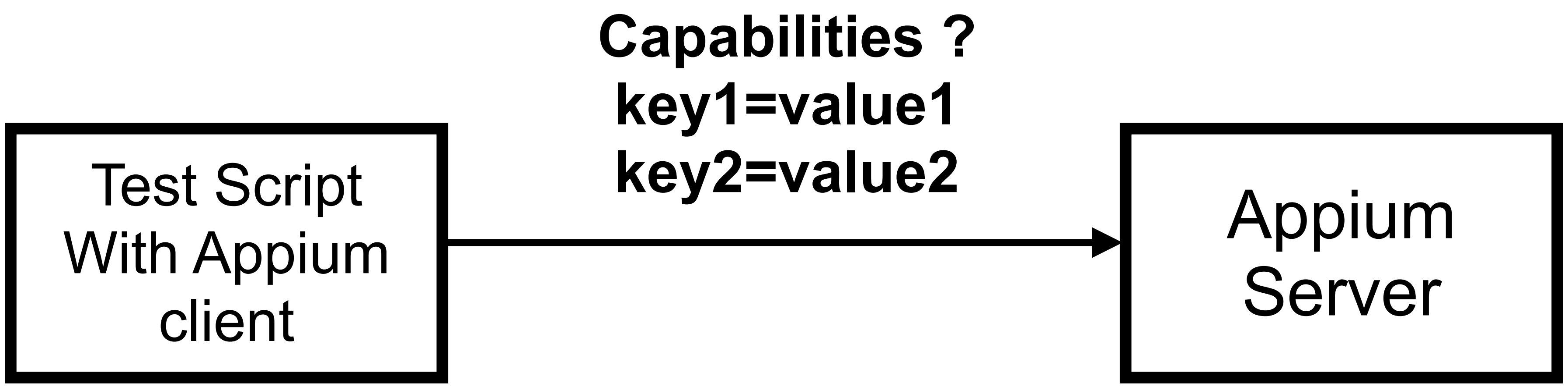
Appium Inspector

GUI inspector for mobile apps



https://github.com/appium/appium-inspector

Create a session with capabilities



http://appium.io/docs/en/writing-running-appium/caps/index.html

Required capabilities

Name	Description
platformName	Platform to automate (iOS, Android)
platformVersion	Version of platform
deviceName	Type of device to automate
app	Path to your app

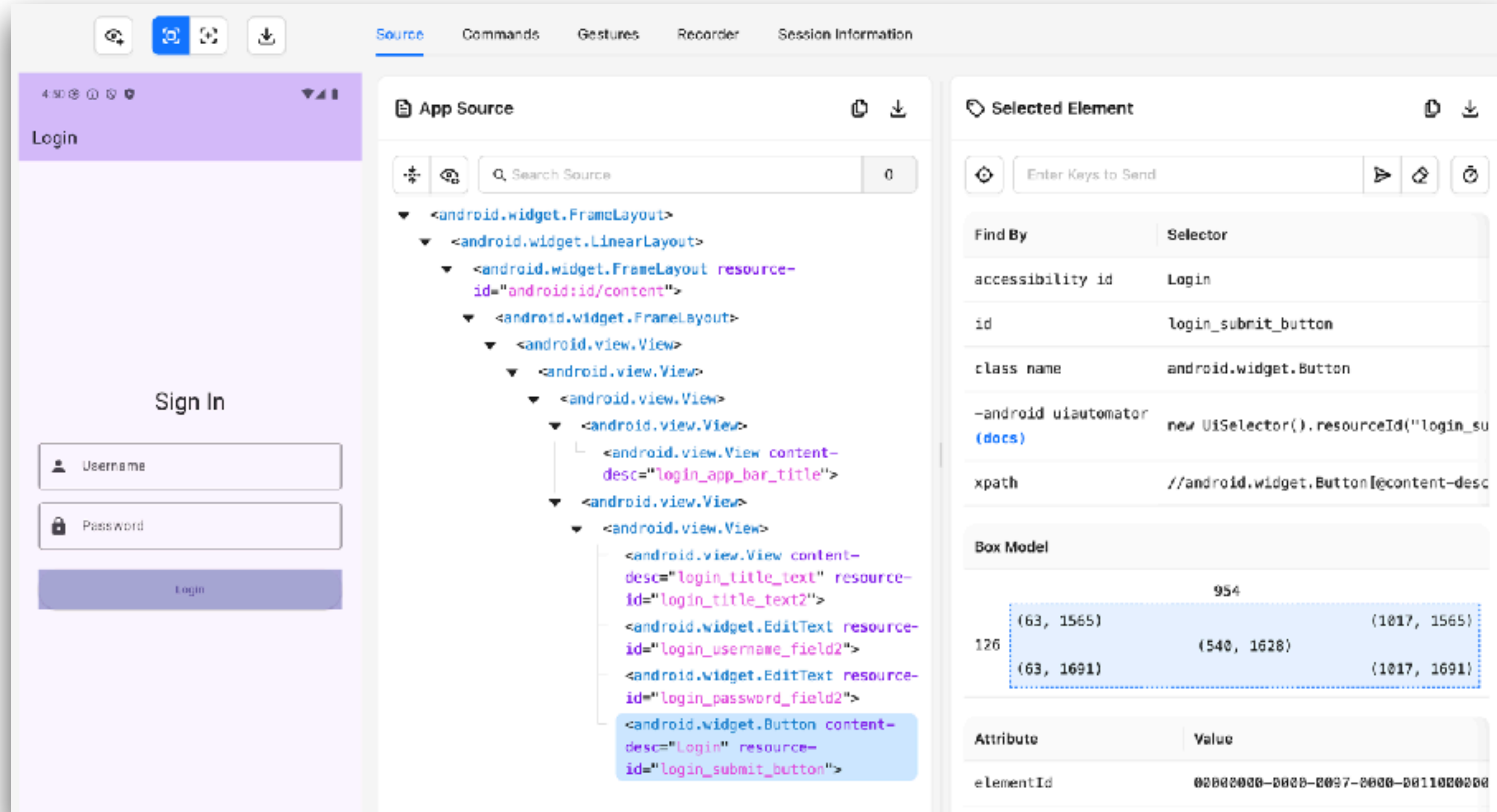
https://appium.io/docs/en/2.1/guides/caps/

Session capabilities for android

```json
{
  "platformName": "Android",
  "appium:automationName": "UiAutomator2",
  "appium:appPackage": "com.demo.registerapp",
  "appium:deviceName": "R58M36QKSJK"
}
```

https://github.com/appium/appium-uiautomator2-driver

Appium Inspector (Android)

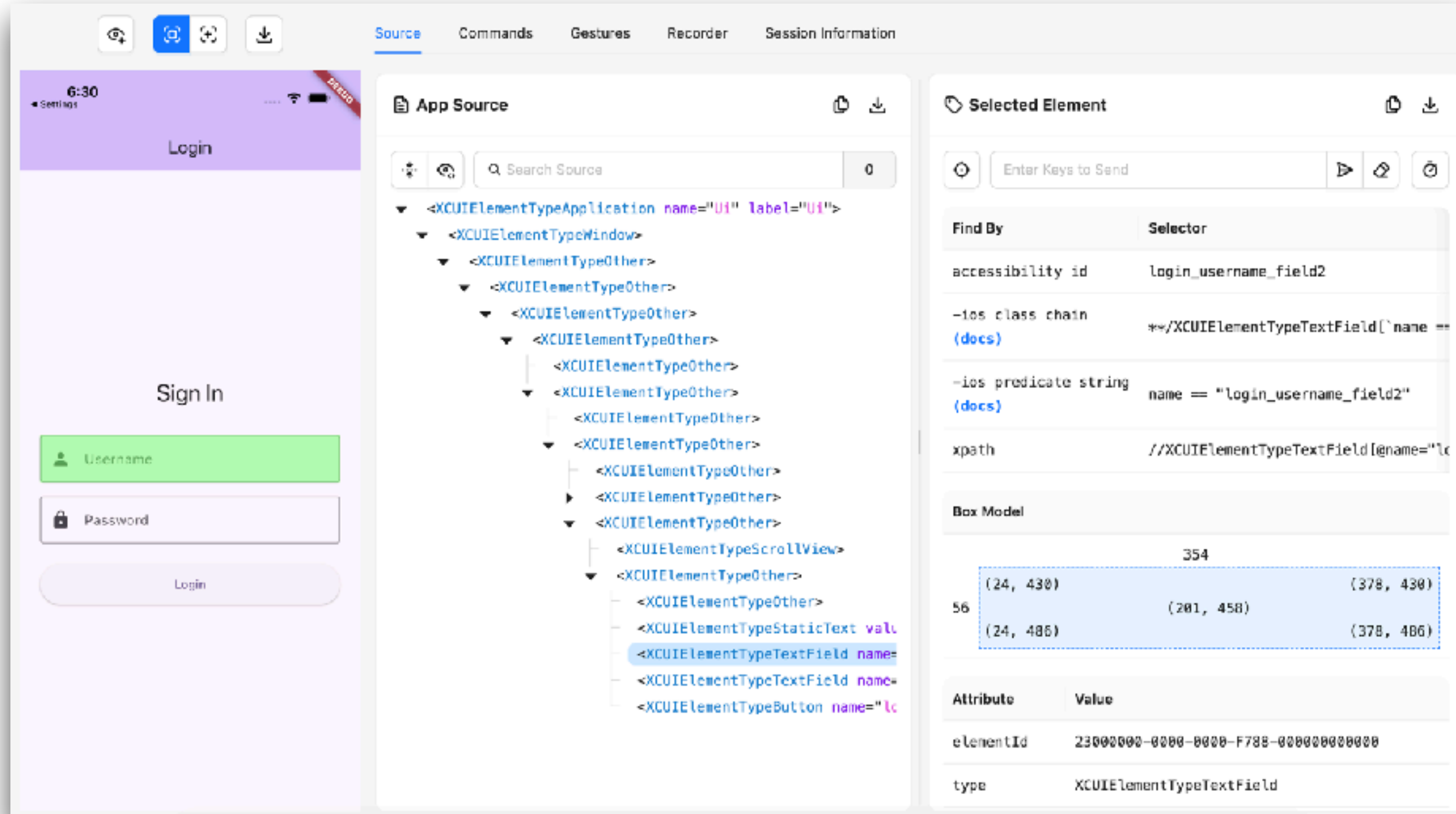


Session capabilities for iOS

```json
{
  "platformName": "iOS",
  "appium:deviceName": "iPhone 17",
  "appium:platformVersion": "26.2",
  "appium:automationName": "XCUITest",
  "appium:bundleId": "com.example.ui",
  "appium:connectionRetryTimeout": 60000,
  "appium:noReset": true
}
```

https://appium.github.io/appium-xcuitest-driver/latest/

Appium Inspector (iOS)



Appium with Flutter App

Android App

iOS App

Finding and Using Elements

Isolated from UI !!

Android App

iOS App

Locator Strategies

Strategy	Description
Accessibility ID	iOS = accessibility-id Android = content-desc
ID	iOS = name Android = resource-id
Name	Name of element
XPath	Pattern of element in XML (not recommended) Absolute path vs Relative path ?
Android UIAutomator	Android only
iOS Predicate String	iOS only

Good Locators

Unique
Descriptive
Resilient
Shorter in length (maintainability)

50 SHADES OF GREEN

Flutter App Locator

Key (default)
Text
Semantic identifier
Tooltip
Type

Working with Semantics widget

Use identifier in Flutter 3.19+

Android App

iOS App

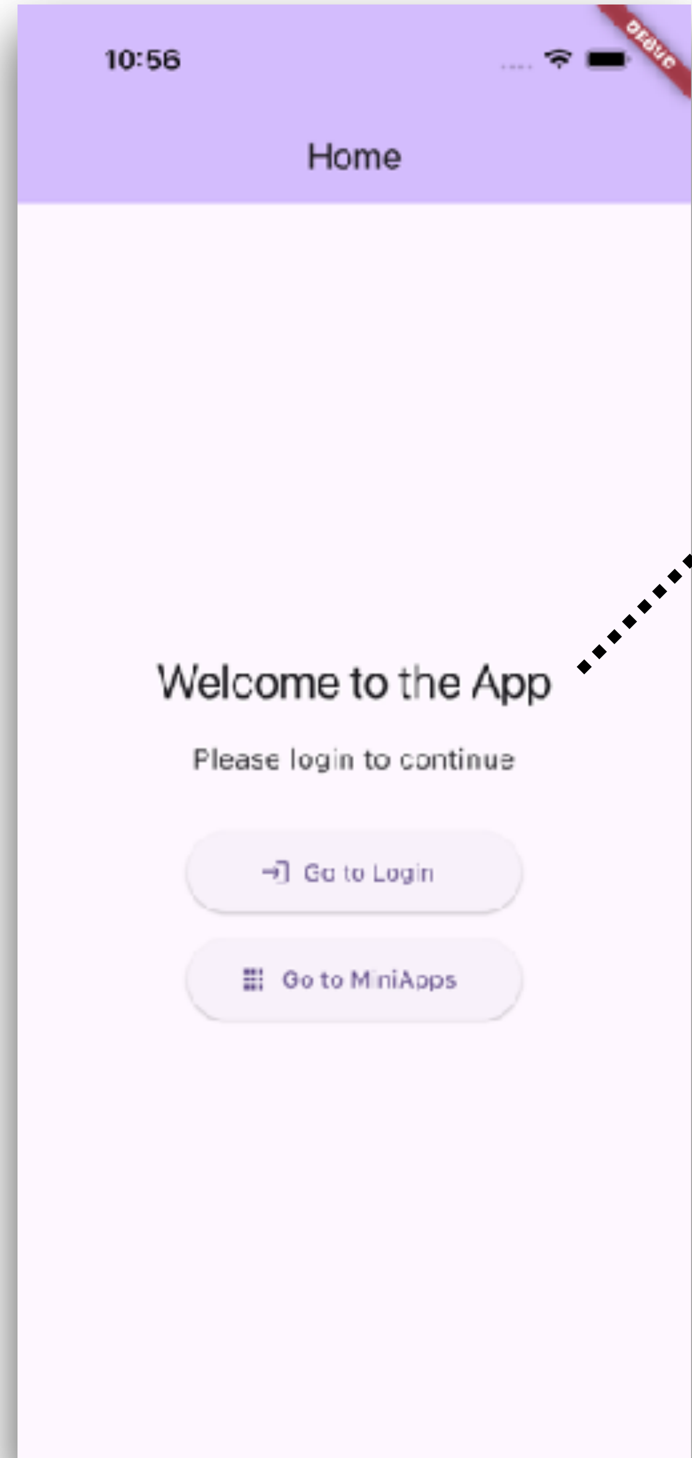
https://blog.flutter.dev/whats-new-in-flutter-3-19-58b1aae242d2

SemanticsProperties accessibility identifier

A new accessibility identifier in `SemanticsProperties` provides an identifier for the semantic node in the native accessibility hierarchy. On Android, it appears in the accessibility hierarchy as `resource-id`. On iOS, this sets `UIAccessibilityElement.accessibilityIdentifier`. We want to thank community member @bartekpacia for this change, which spanned the engine and framework.

https://blog.flutter.dev/whats-new-in-flutter-3-19-58b1aae242d2

Text (iOS app)



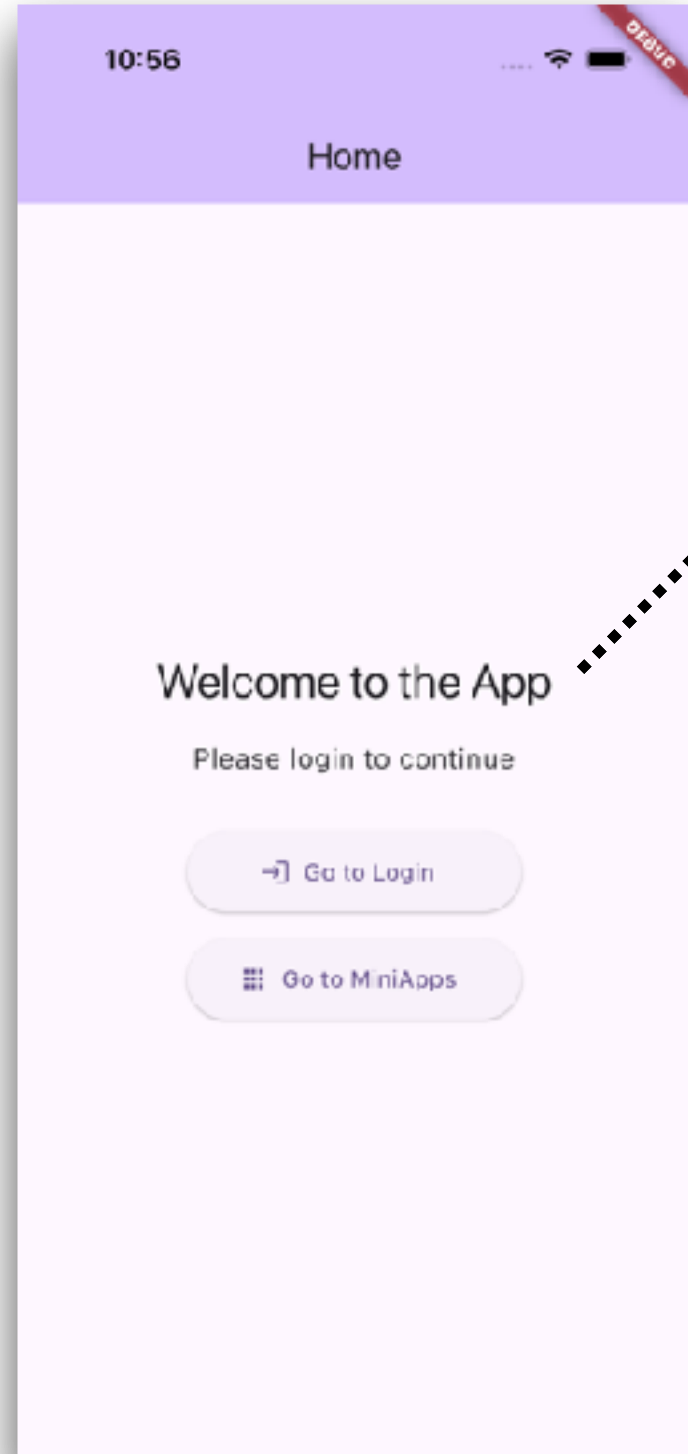
Flutter code

```
Text(
    'Welcome to the App',
    key: const Key('home_welcome_text'),
    semanticsIdentifier: 'home_welcome_text2',
)
```

Appium inspector (accessibility, text)

Find By	Selector
accessibility id	home_welcome_text2
-ios class chain (docs)	**/XCUIElementTypeStaticText[`name =
-ios predicate string (docs)	name == "home_welcome_text2"
xpath	//XCUIElementTypeStaticText[@name="h

Text (Android app)



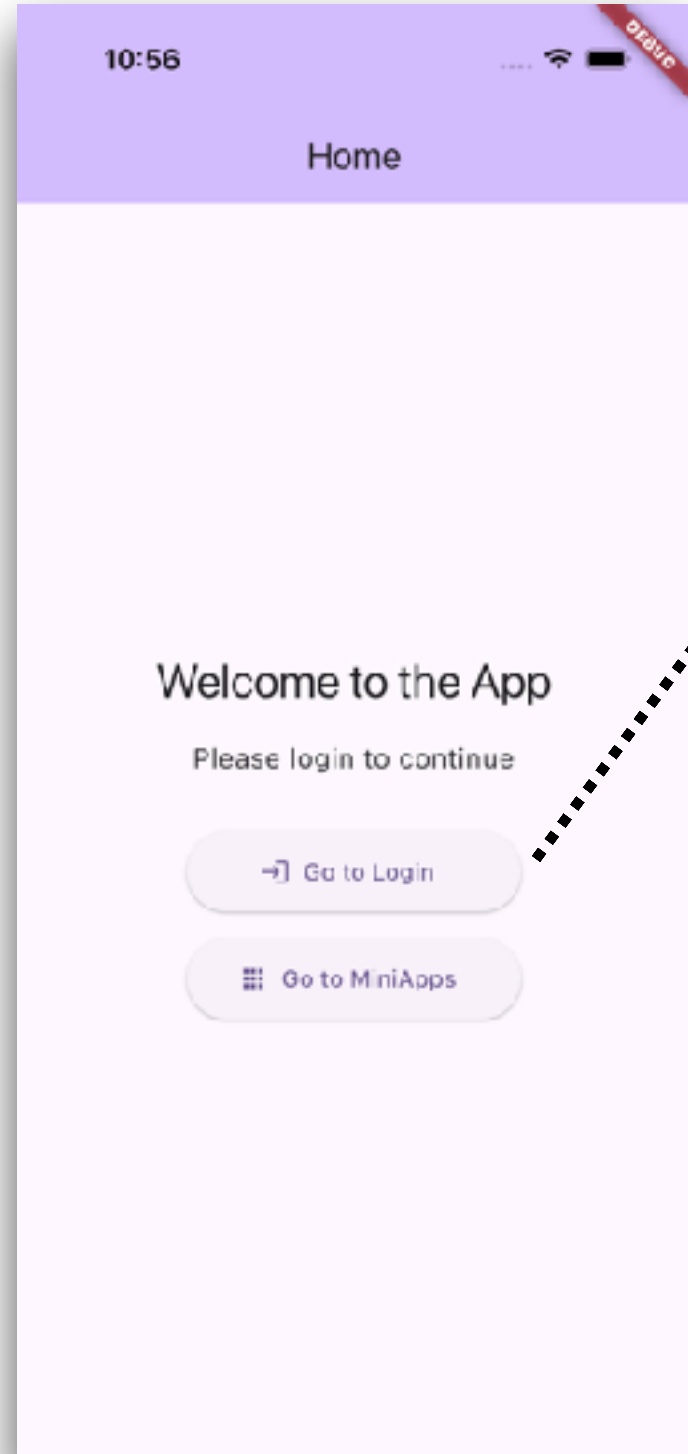
Flutter code

```
Text(
    'Welcome to the App',
    key: const Key('home_welcome_text'),
    semanticsIdentifier: 'home_welcome_text2',
)
```

Appium inspector (resource-id, content-desc)

Find By	Selector
accessibility id	Welcome to the App
id	home_welcome_text2
-android uiautomator (docs)	new UiSelector().resourceId("home_wel
xpath	//android.view.View[@content-desc="We

Button (iOS app)



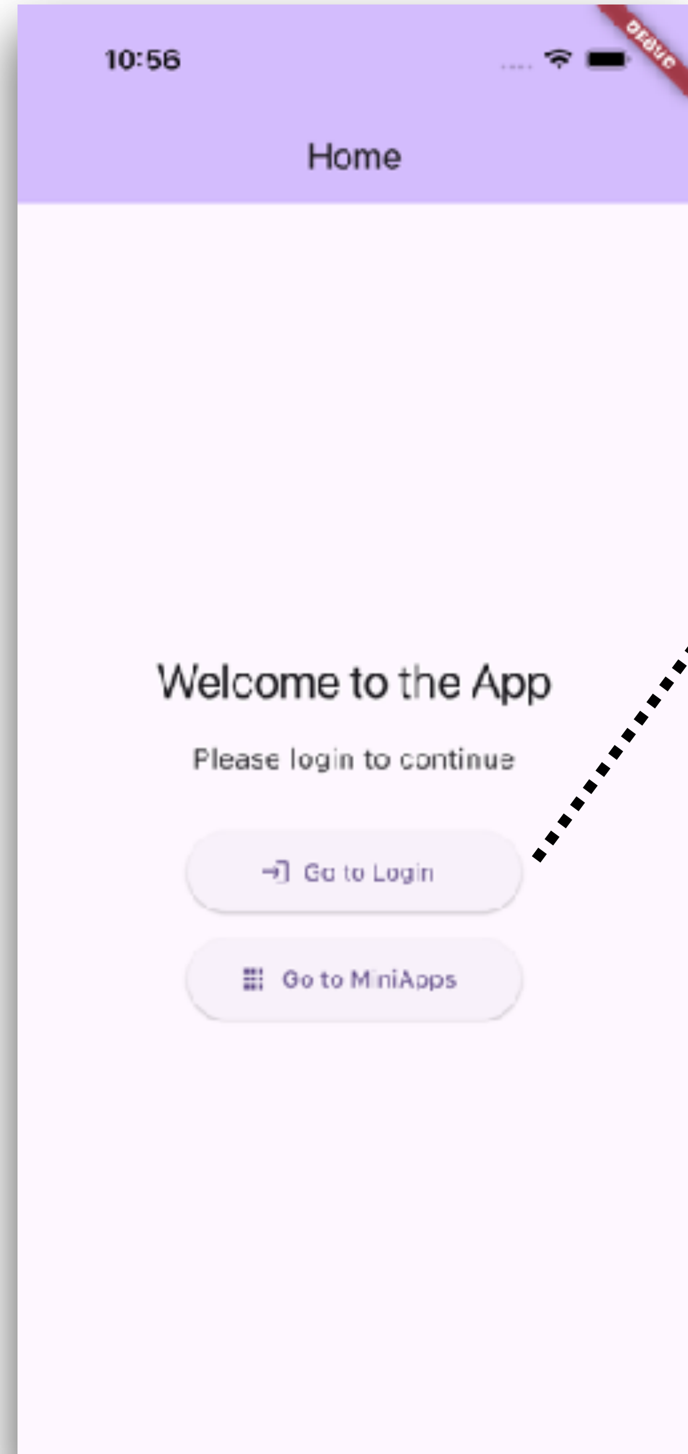
Flutter code

```
Semantics(
  identifier: 'home_login_button',
  label: 'Go to Login',
  excludeSemantics: true,
  child: ElevatedButton.icon(
    key: const Key('home_login_button'),
    label: const Text('Go to Login'),
  ),
)
```

Appium inspector (accessibility, value)

Find By	Selector
accessibility id	home_login_button
-ios class chain (docs)	**/XCUIElementTypeStaticText[`name =
-ios predicate string (docs)	name == "home_login_button"
xpath	//XCUIElementTypeStaticText[@name="h

Button (Android app)



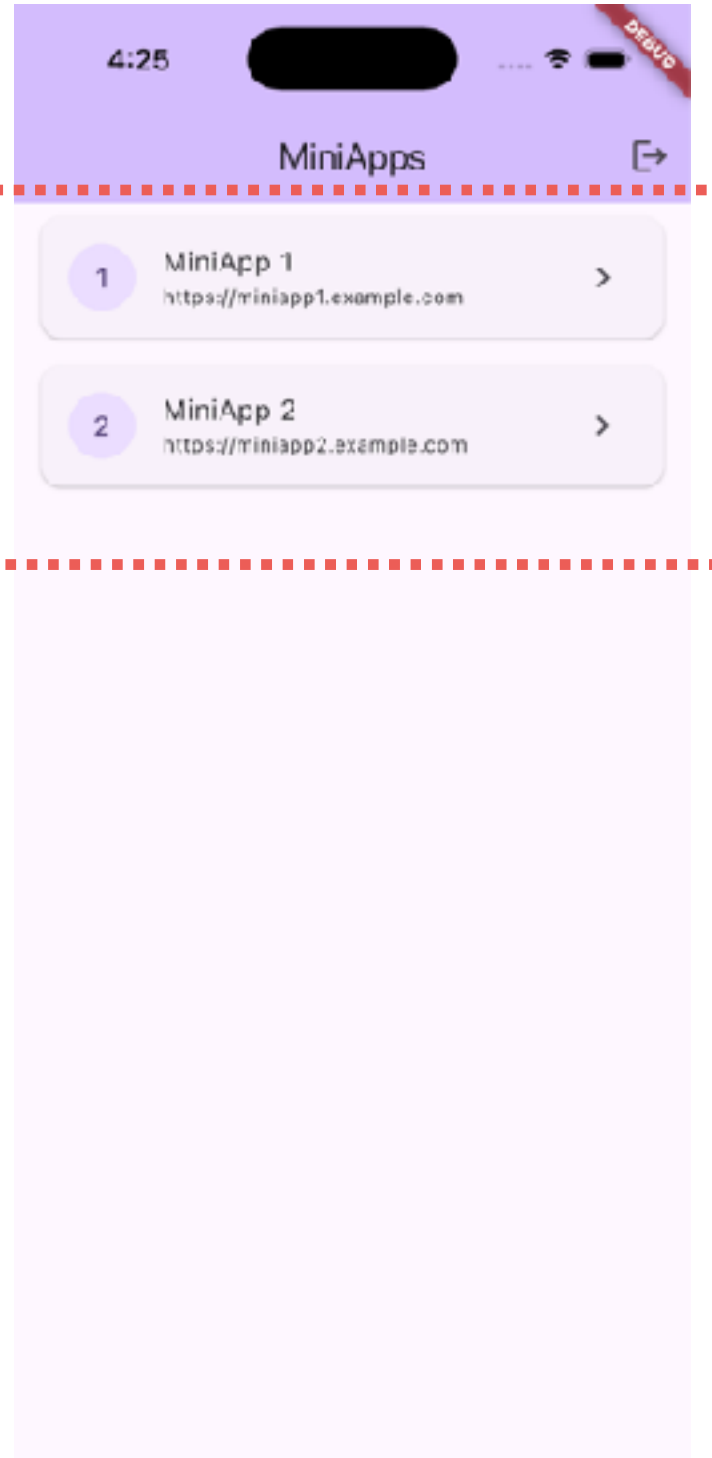
Flutter code

```dart
Semantics(
  identifier: 'home_login_button',
  label: 'Go to Login',
  excludeSemantics: true,
  child: ElevatedButton.icon(
    key: const Key('home_login_button'),
    label: const Text('Go to Login'),
  ),
)
```

Appium inspector (resource-id, content-desc)

Find By	Selector
accessibility id	Go to Login
id	home_login_button
-android uiautomator (docs)	new UiSelector().resourceId("home_log
xpath	//android.view.View[@content-desc="Go

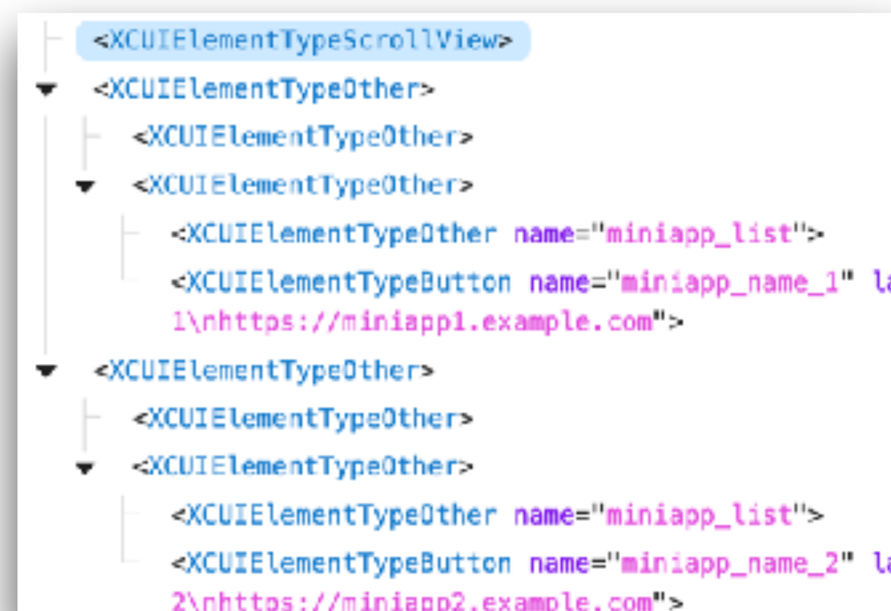
ListView (iOS app)



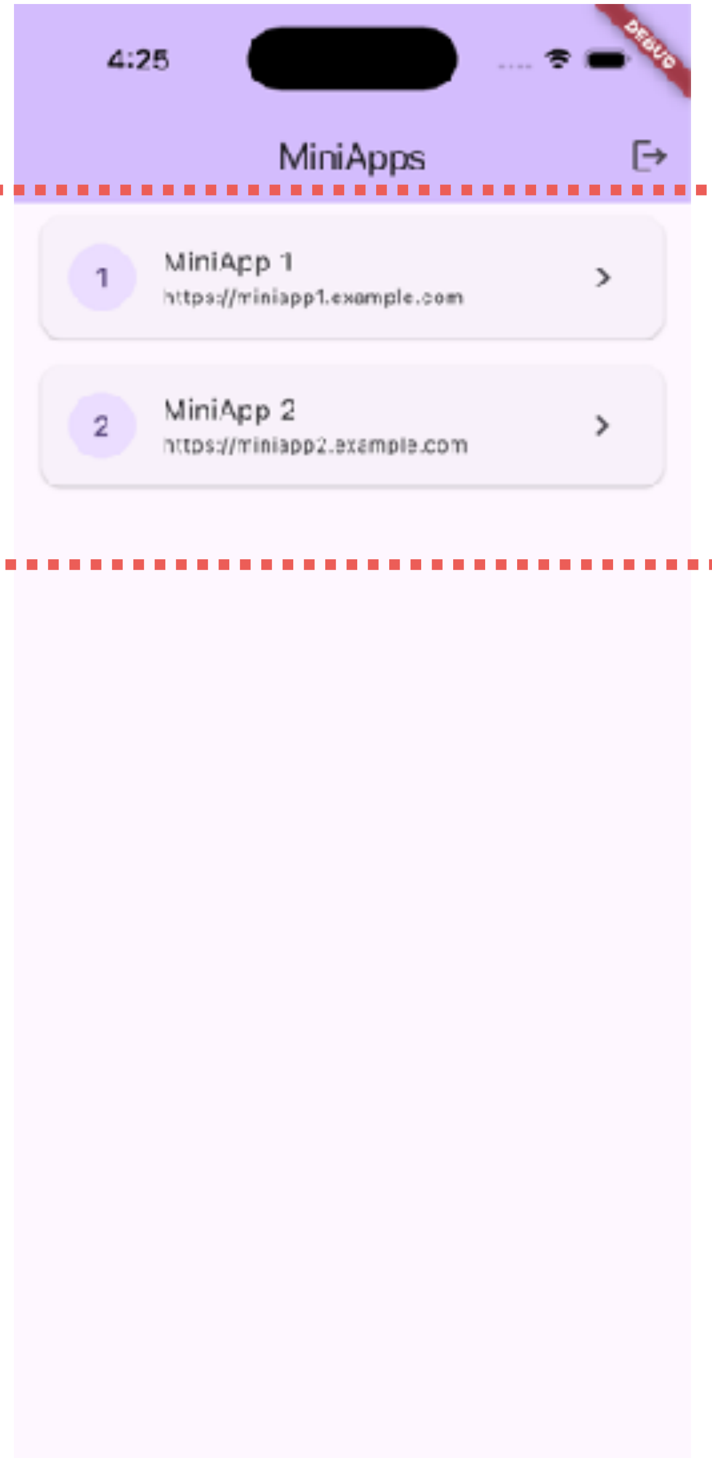
Flutter code

```
return ListView.builder(
  key: const Key('miniapps_list'),
  itemCount: _miniApps!.length,
  itemBuilder: (context, index) {
    final miniApp = _miniApps![index];
    return Semantics(
      identifier: 'miniapp_list',
      container: true,
      child: Card(
        key: Key('miniapp_card_${miniApp.id}'),
```

Appium inspector (accessibility)



ListView (Android app)



Flutter code

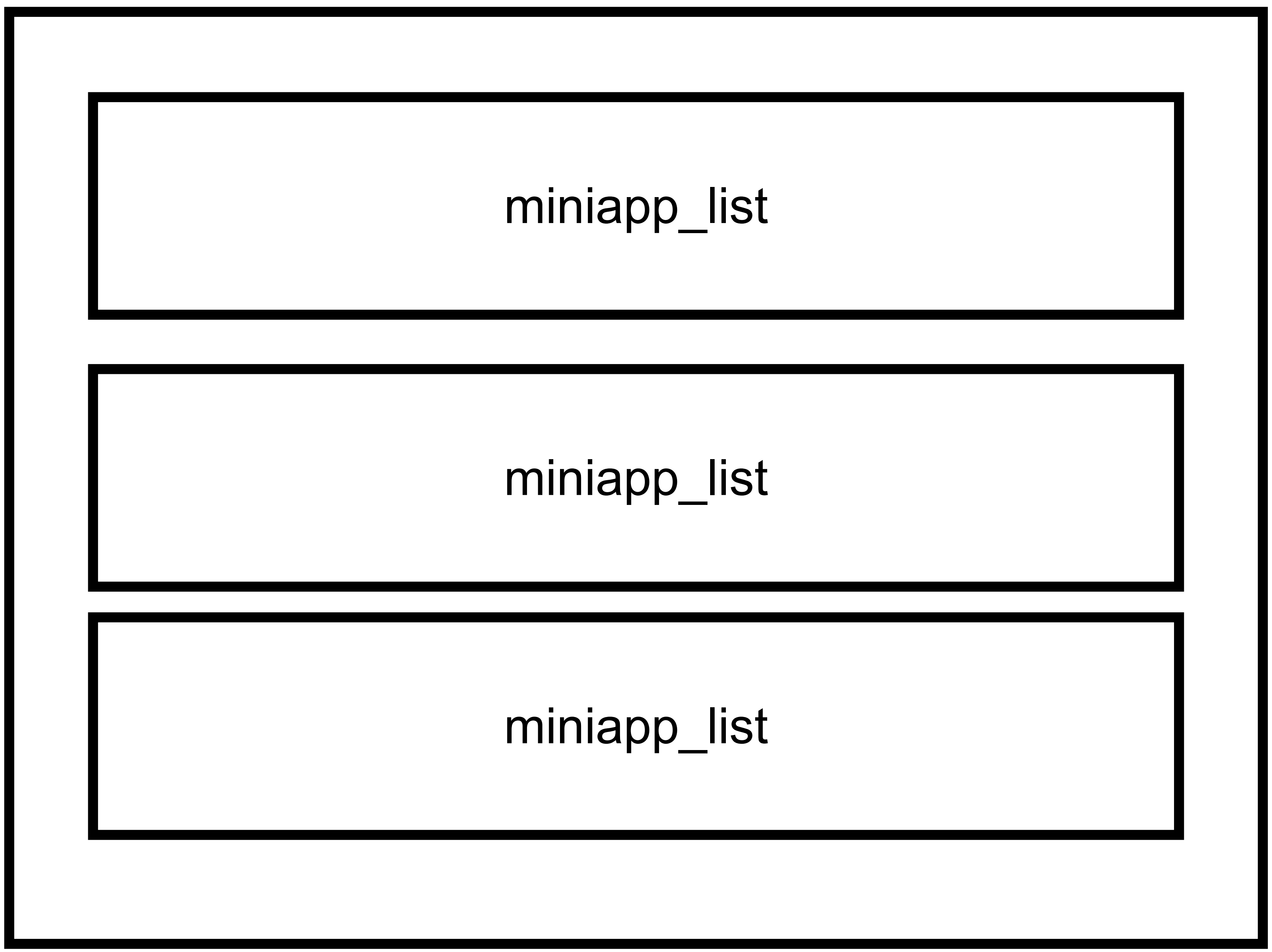
```
return ListView.builder(
  key: const Key('miniapps_list'),
  itemCount: _miniApps!.length,
  itemBuilder: (context, index) {
    final miniApp = _miniApps![index];
    return Semantics(
      identifier: 'miniapp_list',
      container: true,
      child: Card(
        key: Key('miniapp_card_${miniApp.id}'),
```

Appium inspector (resource-id)

```
<android.view.View>
  <android.view.View resource-id="miniapp_list">
    <android.widget.Button content-desc="1\nMiniApp
    1\nhttps://miniapp1.example.com" resource-
    id="miniapp_name_1">
<android.view.View>
  <android.view.View resource-id="miniapp_list">
```

ListView

Size = 3



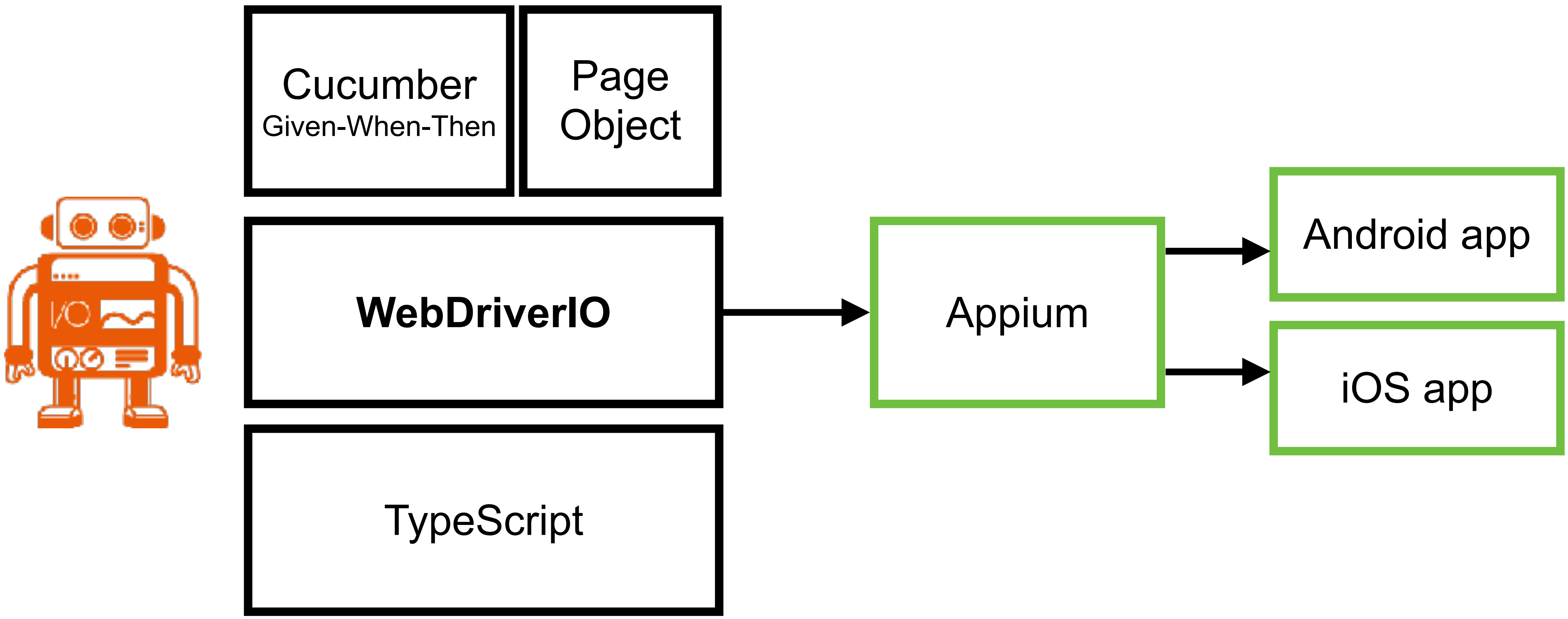
Locator and Value ?

Flutter semantic	iOS	Android
Text	Accessibility-id Text	Resource-id Content-desc
Button	Accessibility-id Text	Resource-id Content-desc
TextField	Accessibility-id Value	Resource-id Value
ListView	Accessibility-id	Resource-id

Write Test Cases

Create Test Project

$npm init wdio@latest .

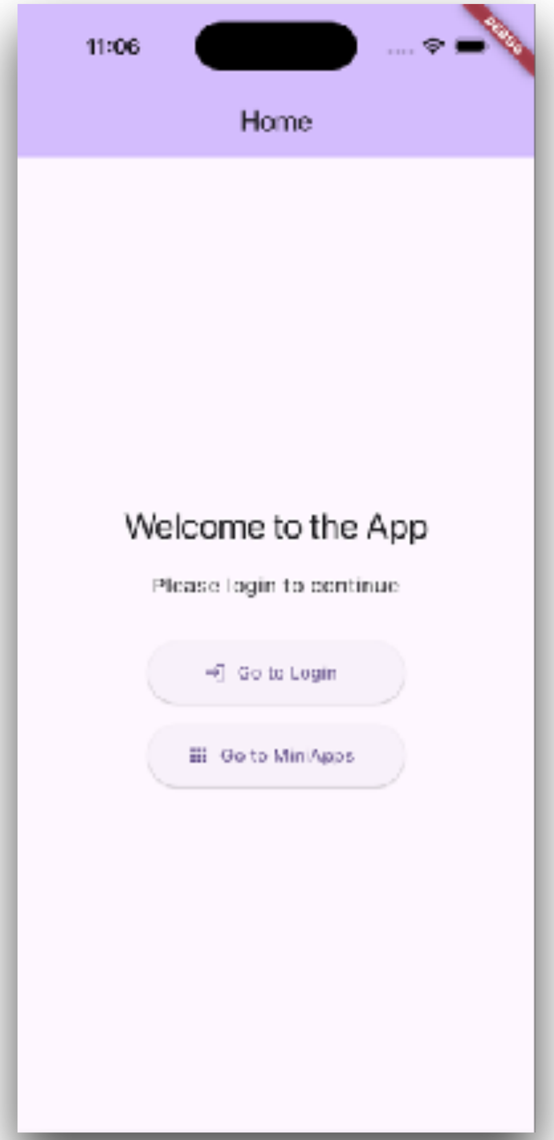


https://webdriver.io/docs/gettingstarted

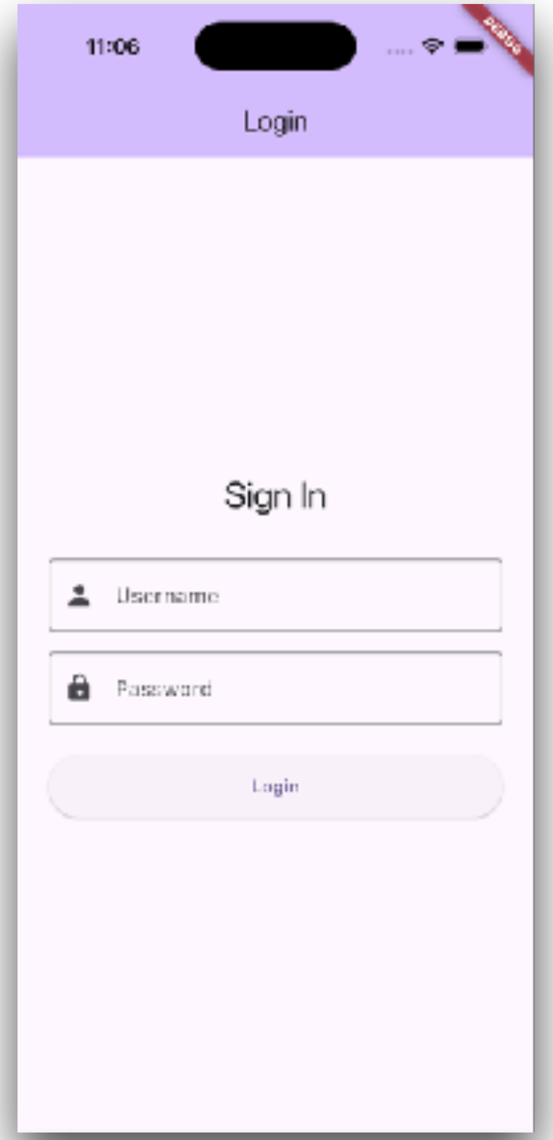
Mobile App for Testing

Flow of Mobile app (1)

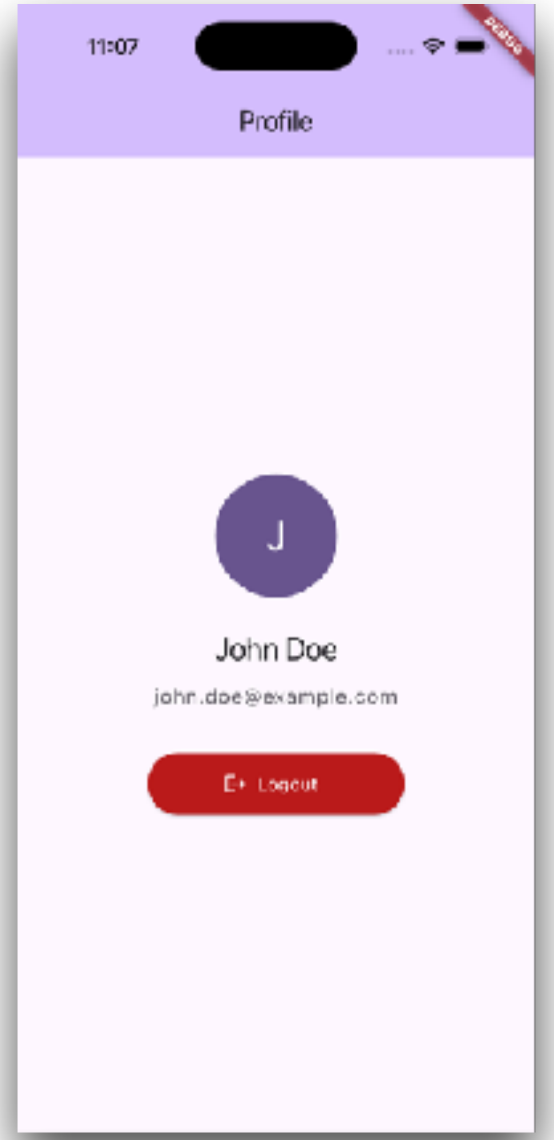
Home



Login

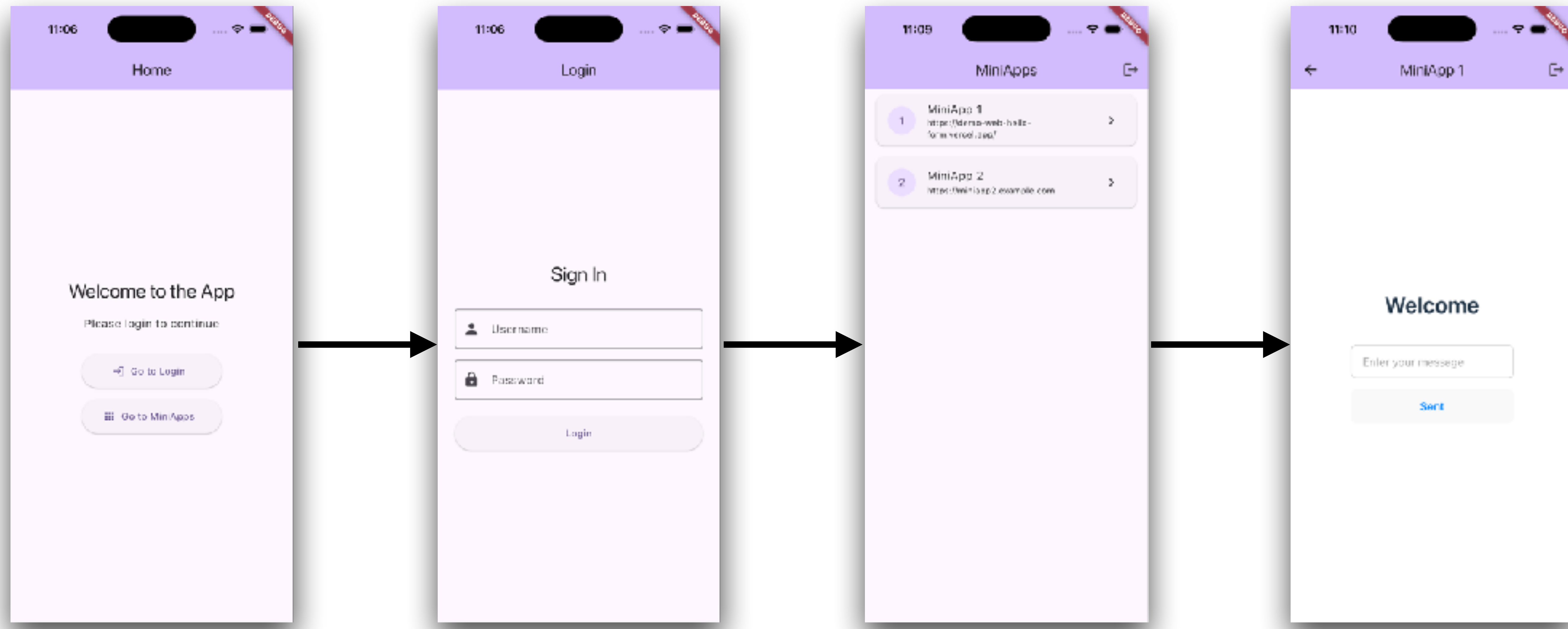


Profile

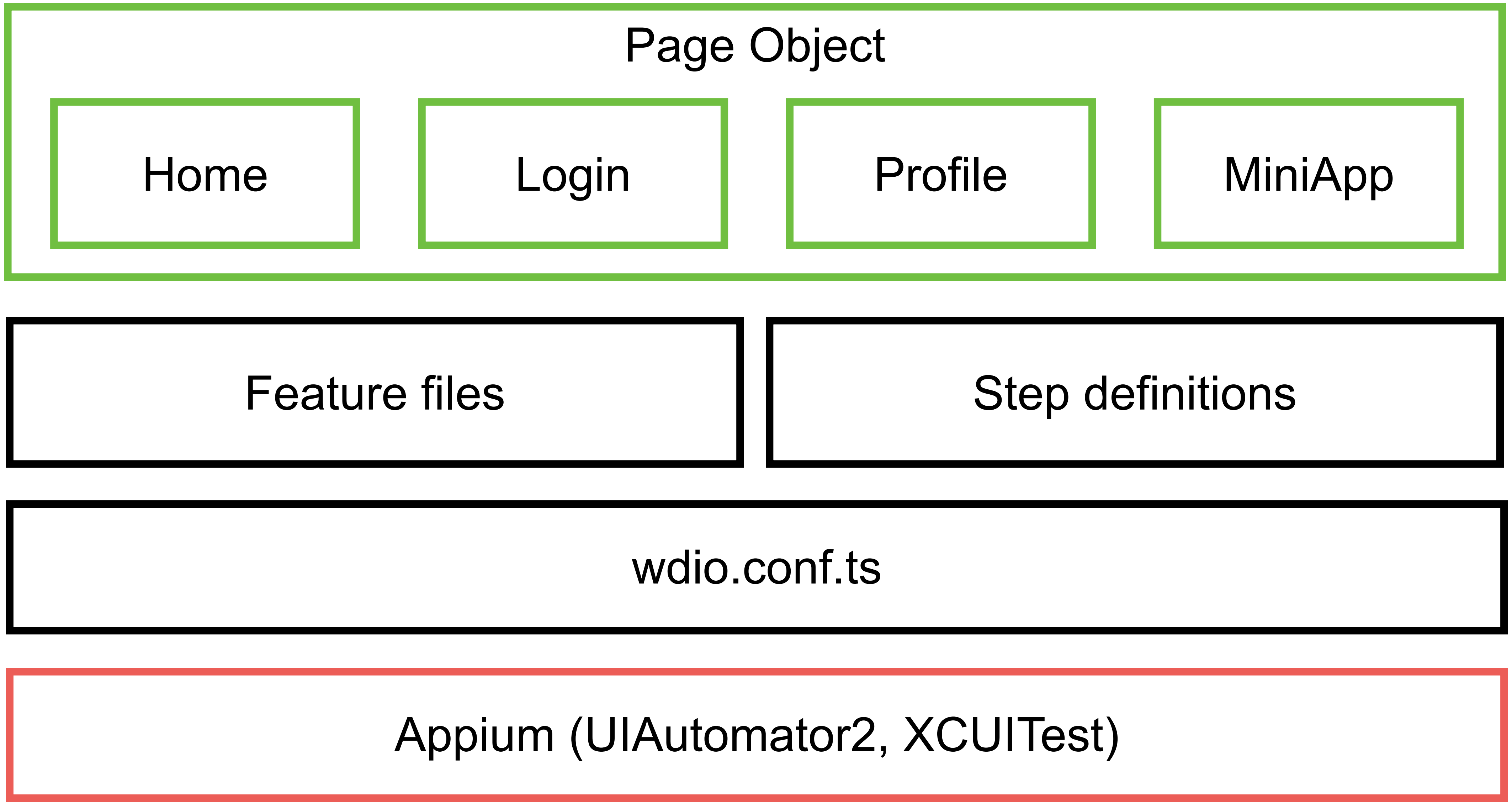


Flow of Mobile app (2)

Home Login MiniAppList WebView



Test Project Structure



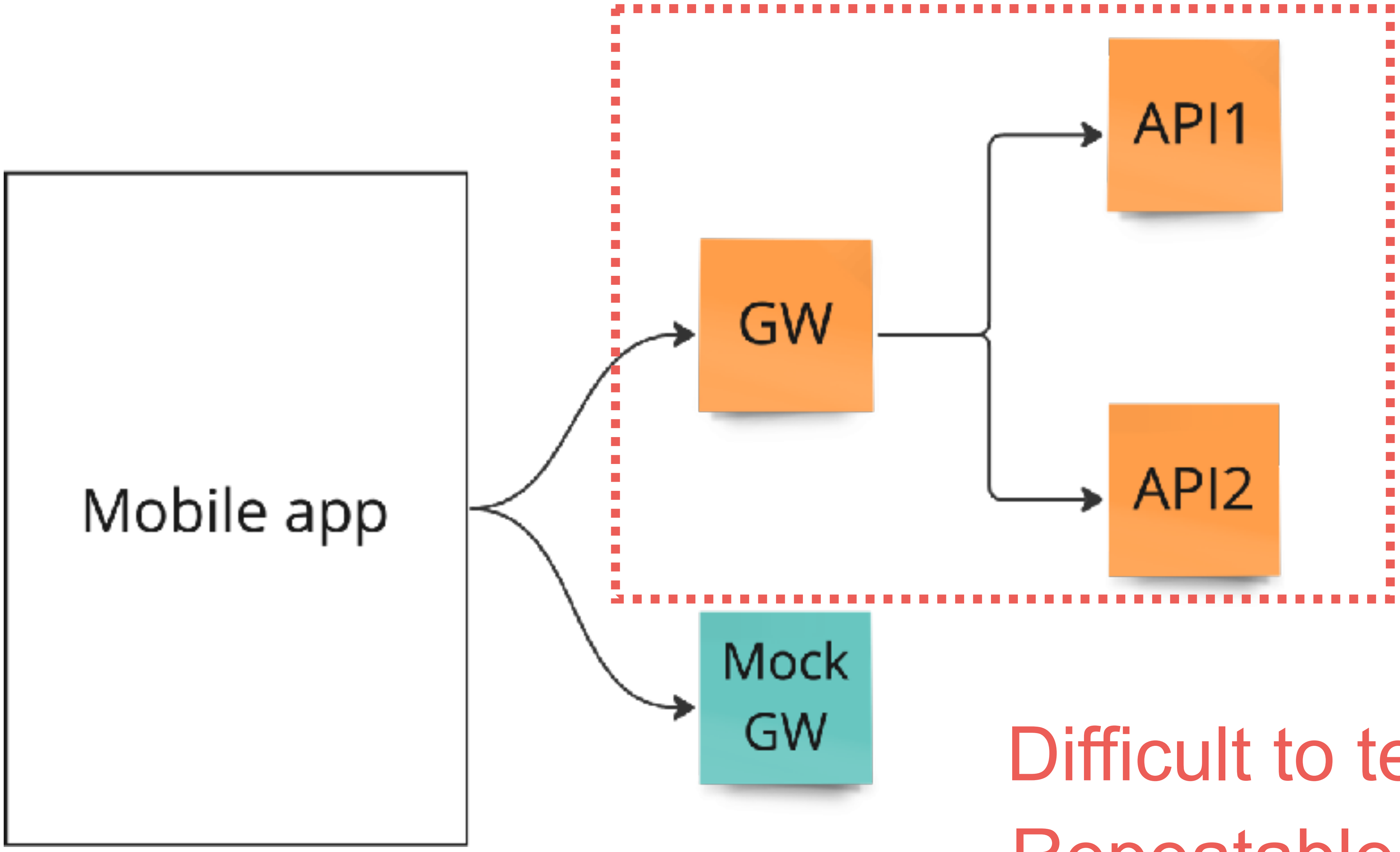
https://github.com/up1/workshop-mobile-testing-with-typescript/tree/main/appium_testing

Test strategy ?

Good Tests ?

Fast
Isolate
Repeatable
Self-verify
Timely

Integration Testing



Difficult to test
Repeatable
Reliable

Q/A